

Varunprasaath R. Selvaraju, Ph.D.

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PROFESSIONAL SUMMARY

PhD chemist/materials scientist with 6+ years diagnosing degradation and failure modes in functional polymer and thin-film systems through multi-technique characterization, DOE/SPC, and root-cause analysis. Translates lab evidence into reproducible test methods, SOPs, and cross-functional engineering recommendations for high-reliability materials.

CORE COMPETENCIES

Materials Failure Analysis

Root-Cause Investigation

Degradation / Contamination Mechanisms

Thin Films, Coatings & Polymers

Spectroscopy & Thermal Analysis

Electrochemical Characterization

DOE / SPC / Test Method Development

Technical Reports & Cross-Functional Communication

WORK EXPERIENCE

University of Colorado Boulder

July 2025 - Present

Postdoctoral Researcher

- Lead DOE-guided process-development experiments for bio-based polymer systems, linking process parameters to material quality, degradation behavior, and reproducible analytical specifications.
- Use FT-IR, spectroscopy, thermal analysis, and statistical process-control concepts to investigate structure-property changes and recommend process/test-method mitigations.
- Translate experimental findings into SOP-ready protocols, calibration expectations, and technical recommendations for repeatable execution across lab teams.

University of Colorado Boulder

July 2021 - July 2025

Graduate Researcher

- Designed and characterized 15+ electrochemical/thin-film material variants using cyclic voltammetry, spectroelectrochemistry, UV-Vis-NIR, FT-IR, NMR, GPC, TGA, and DSC to diagnose structure-property-performance relationships.
- Identified redox instability, oxidative degradation, and interface-related failure modes under electrochemical stress, then recommended material and process mitigations.
- Built reproducible analytical datasets and technical reports supporting root-cause analysis, method comparison, and cross-functional research decisions.

Georgia Institute of Technology

Aug 2019 - July 2021

Graduate Researcher

- Co-led AFOSR-funded process development for high-performance molecular thin-film coatings for defense optical applications, developing surface preparation protocols and CVD-based deposition strategies to achieve controlled film morphology and device performance.

ETH Zurich

May 2017 - July 2017

Research Assistant

- Developed and optimized CVD-based polymer thin-film deposition processes using DOE strategies and in-situ QCM-D process monitoring, establishing process control parameters and characterizing film growth kinetics to improve coating reproducibility and uniformity.

University of Colorado Boulder

2021 - 2025

Laboratory Operations Manager

- Managed shared laboratory operations, instrument access, safety practices, and troubleshooting workflows for multidisciplinary materials-chemistry teams.
- Improved documentation, training, and operating discipline for repeatable measurements and safer high-throughput experimental work.

Georgia Tech

2019-2021

Teaching Assistant, Organic Chemistry I & Lab

- Mentored 120+ undergraduates; developed supplemental materials improving exam performance by 15%.

KEY PROJECTS

Electrochemical Materials Failure-Mode Analysis Root Cause

Diagnosed redox instability, oxidative degradation, and interface-related performance loss in functional electrochemical/thin-film materials, then translated findings into material and process mitigations.

Cyclic voltammetry, spectroelectrochemistry, UV-Vis-NIR, FT-IR, DOE, degradation analysis

Process-Controlled Polymer Systems Process Development

Led DOE-guided process optimization for polymer systems, defining critical parameters, analytical readouts, and reproducible process specifications tied to material quality.

DOE, SPC, FT-IR, thermal analysis, SOPs, root-cause documentation

Thin-Film Coating Process Development Semiconductor-adjacent Materials

Developed CVD/surface-preparation strategies for molecular coatings, translating process variables into film morphology and optical/material performance.

CVD, QCM-D, spectroscopy, surface preparation, process-structure-property analysis

EDUCATION

Ph.D. in Chemistry - University of Colorado Boulder

2025

Dissertation: Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials

Advisor: Prof. Seth Marder

(Transferred from Georgia Institute of Technology)

B.S. (Research), Major: Chemistry; Minor: Materials Science - Indian Institute of Science (IISc)

2019

INSPIRE Scholarship for Higher Education (Top 1% nationally)

CERTIFICATIONS

Statistical Process Control (SPC) - Coursera, 2026

TECHNICAL SKILLS

Failure & Reliability Analysis: Failure mode identification, root-cause analysis, degradation/stability testing, physics-of-failure mindset, FMEA principles, corrective-action recommendations, structured technical reports

Materials Characterization: FT-IR, UV-Vis-NIR, ellipsometry, cyclic voltammetry, spectroelectrochemistry, TGA, DSC, NMR, GPC, QCM-D, GC-MS, LC-MS exposure; cross-technique interpretation of materials data

Materials Systems: Organic/organometallic thin films, electrochromic materials, polymer systems, coatings, surface functionalization, process-structure-property relationships

Data & Quality: DOE, SPC, JMP/basic, Python/basic, Excel, SOPs, calibration expectations, reproducibility studies, lab safety and operations management

JOURNAL PUBLICATIONS

1. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Narrow-Band Electrochromism of a Ferrocene-Substituted Rhodamine B." Chemistry of Materials, 2025, 37, 5558-5568.

2. Mandal, J.; Ramasamy Selvaraju, V.; Yan, W.; Divandari, M.; Spencer, N. D.; Dubner, M. "In Situ Monitoring of SI-ATRP Throughout Multiple Reinitiations Under Flow by Means of a Quartz Crystal Microbalance." RSC Advances, 2018, 8, 20048-20055.
3. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Beyond Rhodamine B: Investigating Alternative Xanthene Cores for Modular Electrochromic Systems." (Manuscript under preparation).

CONFERENCE PRESENTATIONS

Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials, ACS Fall National Meeting, Denver, CO, August 2024.

Electrochemical and Spectroscopic Properties of Substituted Rhodamine Derivatives, Gerischer-Lewerenz Electrochemistry Symposium, Fort Collins, CO, August 2024.

Decoupling Redox and Color for High-Contrast Electrochromics, Excited State Processes Conference, Santa Fe, NM, June 2023.

Electrochromism: Decoupling Color and Redox in Rhodamine-Based Systems, Rocky Mountain Solid State Chemistry Workshop, Boulder, CO, January 2023.

HONORS & AWARDS

Plastic Sustainability Innovation in Material Award | INSPIRE Scholarship (Top 1%, India, 2015) | 97.12 Percentile - IIT JEE Mains (2015) | All India Rank 144 - NEST (2014)